MIL-P-46067B(MR)
9 February 1977
SUPERSEDING
MIL-P-46067A(MR)
1 March 1971

MILITARY SPECIFICATION

PLASTIC EMBEDDING COMPOUND, EPOXY RESIN SYSTEM

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers epoxy resins used as embedding compounds.

1.2 Classification. The epoxy resins shall be of the following compositions, as specified (see 6.2).

Composition 1 – Unfilled
Composition 2 – Silica filled, white
Composition 3 – Silica filled, black

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-C-96 - Can, Metal, 28 Gage and Lighter
PPP-D-705 – Drum, Shipping and Storage, Steel, 16 and 30 Gallon Capacity
PPP-D-729 - Drum, Shipping and Storage, Steel, 55 Gallon
PPP-P-704 – Pail, Shipping, Steel (1 Through 12 Gallon)

FSC 9330,5970

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY
MIL-STD-105 – Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 – Marking for Shipment and Storage
MIL-STD-147 – Palletized and Containerized Unit Loads 40 Inch X 48 Inch Pallets, Skids, Runners or Pallet Type Base

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 257 - D-C Resistance or Conductance of Insulating Materials
D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing
D 790 - Flexural Properties of Plastics
D 792 - Specific Gravity and Density of Plastics by Displacement
D 1652 – Epoxy Content of Epoxy Resins
D 2393 – Viscosity of Epoxy Compounds and Related Components
D 2471 – Gel Time and Peak Exothermic Temperature of Reacting Thermosetting Plastic Compositions
D 2583 – Indentation Hardness of Plastics by Means of a Barcol Impressor
D 2584 – Ignition Loss of Cured Reinforced Resins

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.

Department of Transportation

49 CFR 171-189 – Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Department of Transportation Regulations are now a part of the Code of Federal Regulations (latest cumulative packet supplement) available from the Superintendent of Documents, Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "Code of Federal Regulations, 49 CFR 171-189".)

American National Standards Institute, Inc.

ANSI K 68.1 – Guide for Classifying and Labeling Epoxy Products According to their Hazardous Potentialities.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The material shall consist of epoxy resin with the necessary diluents and fillers (as applicable), required to conform to this specification.

3.2 Property Values. The material shall conform to the property values specified in table I, when tested as specified in the applicable procedure of 4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.

Table I. Property values

Property	Value		
	Composition 1	Composition 2	Composition 3
Liquid without hardener:			
Viscosity at 25° ± 0.5°C. (77° ± 0.9°F.), centipoises			
Min.	500	9,000	9,000
Max.	900	22,000	22,000
Weight per epoxy equivalent	1/	1/	1/
After casting:			
Specific gravity			
Min.	1.13	1.59	1.59
Max.	1.23	1.69	1.69
Hardness, Barcol units			
Min.	20	50	50
Max.	40	60	60
Ignition loss, percent			
Min.	N.A.2/	48	48
Max.	N.A.2/	52	52
Insulation resistance, Megohms, min.	10^5	10^5	10^5
Flexural strength, 3/ MPa (psi) min.	103 (15,000)	68.9 (10,000)	68.9 (10,000)
Tangent modulus of elasticity, 3/ MPa (psi) min.	2760 (0.4 x10^6)	6210 (0.9 x10^6)	6210 (0.9 x10^6)

1/ The weight per epoxy equivalent shall be agreed upon between supplier and procurement activity, unless a value or range is specified by the procuring activity (see 6.2).

2/ Not applicable.

3/ Metric abbreviation MPa = megapascal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.

3.2.1 Additional property values. When specified by the procuring activity, the contractor shall supply data for shrinkage, dielectric strength, dielectric constant, and volume resistivity of the bulk material supplied to the Government.

3.3 Suitability of resin. The epoxy embedding compound shall be suitable for application and use on electronic assemblies and individual parts. It shall preserve the electrical properties of the equipment to which it is applied by sealing against such environmental conditions as moisture, dirt, fumes, fungus, or other deleterious substances which may be encountered in the military services. It shall be capable of mechanically supporting the part or embedded assembly and shall not require a case or other external support after curing has been completed, though a potting container may be used. It shall not cause deterioration of materials used in electronic assemblies and parts.

3.4 Pot life. The compound shall have a pot life long enough so as to produce uniform castings, free of bubbles and cracks that may affect serviceability. The manufacturer shall furnish instructions on proper mixing of components. When required, the procuring activity shall specify minimum pot life (6.2).

3.5 Workmanship. The epoxy embedding compound shall be a clean homogeneous mixture free from lumps and foreign material. The diluents and fillers shall be thoroughly dispersed in the compound, as determined by visual examination. For compositions 2 and 3, if required, slight stirring may be used to return the material to the homogeneous thoroughly dispersed condition.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, examined, and tested in accordance with the requirements of this specification and, to the extent specified, of all referenced subsidiary specifications and standards. In the event of conflict, this specification shall govern. A supplier's certification of compliance with 3.1 and 3.3 shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in the units of 22.6 kg (50 pounds) for examination in 4.4.2.1.1, and in units of shipping containers for examination in 4.4.2.1.2.

4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination, specified in table II, shall be approximately one pound selected just prior to the closure operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.

Table II. Examination for defects in appearance and workmanship

Examine	Defect
Appearance and workmanship of resin	Not clean
	Not homogeneous
	Not free from lumps and foreign material
	Diluents and fillers not thoroughly dispersed
	Setting of filler material in compositions 2 and 3 during storage

4.2.2.1.2 Examination of packaging. An examination shall be made in accordance with table III to determine that preservation, packing and marking comply with specified contract requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.

Table III. Examination of packaging

Examine	Defect
Preservation	Not level specified; not in accordance with contract requirements
	Cans, pails, or drums not packed as specified
	Packing material not as specified; closures not accomplished by specified or required methods or materials.
Packing	Not level specified; not in accordance with contract requirements.
	Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability
	Inadequate application of components such as: incomplete closures of case liners; container flaps, loose or inadequate strappings, bulged or distorted containers.
Quantity of material	Less than specified or indicated quantity
Weight	Gross weight exceeds specified requirements
Markings	Interior or exterior markings omitted, illegible, incorrect, of improper size, location, sequence, or method of application. Not in accordance with contract requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.

4.2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL) expressed as defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.4.2.1.1	II	2.5
4.4.2.1.2	S-2	2.5

4.2.3 Classification of tests. All tests under this specification shall be classified as lot acceptance tests. They shall be made on each lot of material offered against this specification and, in conjunction with the above examination, shall be the basis of acceptance or rejection of the lot.

4.2.4 Testing. The material shall be tested for the applicable characteristics listed in table I in accordance with the test methods specified herein. The lot size, for the purpose of determining the sample size for testing (see MIL-STD-105) shall be expressed in units of 22.6 kg (50 pounds). The sample unit shall consist of sufficient material to prepare all required specimens. The inspection level shall be S-1, with an acceptance number of 0. The results for each test shall be averaged results of the specimens, except where only one determination is required.

4.3 Test methods.

4.3.1 Viscosity of liquid resin. The viscosity of the liquid resin without hardener shall be determined in accordance with ASTM 2393 at a temperature of 25° ± 0.5°C (77° ± 0.9°F). Conditioning shall be at 25° ± 0.5°C (77° ± 0.9°F) for a minimum of 16 hours prior to testing. The Brookfield RVF viscometer shall employ spindle #1 at 10 revolutions per minute for composition 1, and spindle #4 at 4 revolutions per minute or spindle #5 at 10 revolutions per minute for compositions 2 and 3. One determination shall be made.

4.3.2 Weight per epoxy equivalent. The weight per epoxy equivalent shall be determined in accordance with ASTM D 1652. Two determinations shall be made.

4.3.3 Specimen curing, conditioning, and testing. Specimens of epoxy embedding compound shall be cured at 25° ± 5°C (77° ± 9°F) for a minimum of 7 days, and conditioned in accordance with procedure A of ASTM D 618. Testing of specimens shall be at 23° ± 2°C (73.4° ± 3.6°F) and 50 ± 5 percent relative humidity, except for ash content determination (see 4.5.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.

4.3.4 Specific gravity. Two specimens, of any convenient size, shall be tested in accordance with A-1, A-2, or A-3 of ASTM D 792.

4.3.5 Hardness. The Barcol hardness shall be determined in accordance with ASTM D 2583, using a minimum specimen thickness of 1/8 inch. The minimum number of readings shall be in accordance with ASTM D 2583, using the Model 934-1 instrument.

4.3.6 Ignition loss. Three specimens shall be tested in accordance with ASTM D 2584.

4.3.7 Insulation resistance. Three specimens shall be tested in accordance with ASTM D 257. Prior to testing the specimens shall be conditioned in accordance with procedure C of ASTM D 618, and tested under the conditions of procedure C of ASTM D 618.

4.3.8 Flexural strength and tangent modulus of elasticity. Three specimens shall be tested in accordance with ASTM D 790. Specimen dimensions shall conform to table I of ASTM D 790, using a nominal thickness of 1/8 inch minimum.

4.3.9 Pot life. One determination for pot life shall be made in accordance with ASTM D 2471. Sample size for compositions 2 and 3 shall be 415 ml, and that for composition 1 shall be 120 ml.

5. PACKAGING

Application. The requirements of section 5 apply only to purchase by or direct shipment to the Government.

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. The resin shall be packed as specified (see 6.2), in 1 gallon or smaller cans conforming to type V, class 2, round of PPP-C-96, 5 gallon pails conforming to type II, class 3 of PPP-P-704, 16 gallon drums conforming to type III of PPP-D-705 or 55 gallon drums conforming to type III or IV of PPP-D-729. Cans conforming to PPP-C-96 shall be rust resistant coated on the exterior in accordance with plan B and the side seam shall be striped with a suitable corrosion resistant coating.

5.1.2 Level B. Preservation shall be the same as for level A (see 5.1.1).

5.1.3 Level C. The cans, pails or drums shall be unit packed in accordance with commercial practice to provide adequate protection against deterioration and physical damage during the shipment from the supply source to the first receiving activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Resin packaged in containers conforming to PPP-C-96 shall be packed in accordance with the overseas requirements of the appendix thereto. Resin packaged in containers conforming to PPP-P-704, PPP-P-705 or PPP-D-729 require no overpacking. When specified (see 6.2), 5 gallon pails and 16 gallon drums shall be palletized in accordance with MIL-STD-147.

5.2.2 Level B. Resin packaged in containers conforming to PPP-C-96 shall be packed in accordance with the domestic requirements of the appendix thereto. Resin packaged in containers conforming to PPP-P-704, PPP-P-705 or PPP-D-729 require no overpacking. When specified (see 6.2), 5 gallon pails and 16 gallon drums shall be palletized in accordance with MIL-STD-147.

5.2.3 Level C. The cans, pails or drums shall be packed in accordance with commercial practice adequate to ensure acceptance and delivery by the carrier for the mode of transportation employed. Containers shall comply with the requirements of the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable to the mode of transportation.

5.3 Marking. Interior packages, exterior shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

5.3.1 Hazardous chemicals. All packaged units of hazardous chemicals shall have affixed thereto such warning labels and markings as may be required by the Department of Transportation CFR, Title 49, Part 171-189 and the American National Standards Institute ANSI K 68.1.

5.3.2 Warning markings. All packaged units shall be marked:

> Warning: USE WITH ADEQUATE VENTILATION – MAY BE FLAMMABLE – The solvents contained in this compound may have flash points well below prevailing room temperatures. Do not heat by, or use near, and open flame. Heat preferably by means of a steam coil or other nonflame-producing heat source.
>
> WARNING: Catalyst promoters, and solvent to be used with this compound should be stored in original tight containers isolated from one another in cool locations. Never permit a catalyst to come in contact with any component except the base compound.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The epoxy embedding compounds are intended for use in encasing coils, capacitors and the like, or in assemblies of electrical components or for forming small structural details of an electrical insulating nature.

6.1.1 Composition 1. Composition 1 is used in applications where low viscosity is a special advantage.

6.1.2 Composition 2. Composition 2 is used in preference to Composition 1 when a lower cure shrinkage, a coefficient of thermal expansion closer to that of the ordinary metals, and a higher thermal conductivity are desired.

6.1.3 Composition 3. Composition 3 is used for the same applications as Composition 2 as their properties are identical, except for color.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Composition required (see 1.2).
c. Weight per epoxy equivalent, if a specific value or range is required
d. Property values for shrinkage, dielectric strength, dielectric constant, and volume resistivity (see (3.2.1).
e. Minimum pot life (see 3.5).
f. Level of preservation and packing required (see 5.1, and 5.2).
g. Quantity of cans, pails or drums required, as applicable (see 5.1.1).
h. Whether palletizing is required (see 5.2.1, and 5.2.2).
i. Any special marking required (see 5.3).

6.3 Preparation of compositions. The embedding compounds may be prepared according to the following formulas:

Ingredient	Parts by weight		
	Comp. 1	Comp. 2	Comp. 3
Epoxy resin	100	100	100
Filler	-	125	125
White paste	-	2	-
Black paste	-	-	2
Hardener	20-25	20-25	20-25

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.

6.3.1 Epoxy resin. An epoxy resin which has been found satisfactory has the following properties:

Property	Value	
	Min.	Max.
Specific gravity, 23°C	1.12	1.14
Viscosity, centipoises, 25°C	600	900
Weight per epoxy equivalent	0.52	0.56
Hydrolyzable chlorine, percent	-	0.2

6.3.2 Filler. A filler which has been satisfactory is 325 mesh silica.

6.3.3 White paste. A white paste which has been found satisfactory consists of titanium dioxide dispersed in epoxy resin containing a minimum of 60 percent titanium dioxide by weight.

6.3.4 Black paste. A black paste which has been found satisfactory consists of carbon black dispersed in epoxy resin containing a minimum of 10 percent carbon black by weight.

6.3.5 Hardener. A hardener which has been found satisfactory is a glycidyl polyether-polyamine described in U.S. Patent 2,901,461 having the following properties:

Property	Value	
	Min.	Max.
Specific gravity, 23°C	1.06	1.08
Viscosity, centipoises, 25°C ± 0.5°C	2,500	4,000

Custodian:
Army – MR

Preparing activity:
Army – MR

Review activities:
Army – MI, PA, EL, MD, ME
DSA – GS

Project No. 9330-A527

User activities:
Army - AT

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:46Z
Check the source to verify that this is the current version before use.